

(주)동인엔시스

■ Company Profile

**Your control panel runs
the equipment.
Now it reads it.**

Intelligent Control Panel · Industrial Equipment Data Infrastructure

DONG-IN ENSIS CO., LTD. | September 2026



■ Company overview

After 35 years of building control panels, we turn them into data infrastructure.

Founded in 1991 as a supplier of industrial control equipment, we design and build control systems for ships, energy facilities and plants. Over 750+ projects we learned that most equipment is already controlled by PLCs and drives, yet its data never leaves the control panel.

Our Intelligent Control Panel collects data where the equipment is controlled and delivers it to the cloud in a standard format.

Company	DONG-IN ENSIS CO., LTD.
CEO	Min-gi Baek
Founded	October 1991 (incorporated July 1997)
Business	Intelligent Control Panels; industrial and marine control systems; system integration; power and control products
Sites	Head office & R&D center (Busan); Factories 1 & 2 (Yangsan)
Systems	ISO 9001 · 14001 · 45001
Partners	Schneider Electric official SI partner; ABB · Fuji Electric · Eaton · Danfoss

■ Our journey

From components to systems, from systems to data

35+

Years (since 1991)

750+

Projects delivered

15+

Industries served

40+

CE·UL certified solutions

○ 1991 –

Control equipment supply

Global power and control brands

○ 2006 –

Control system engineering

Design, build, software,
commissioning

○ 2020 –

**Green ship & hydrogen
R&D**

R&D center, national projects, H2
stations

● 2026 –

Intelligent Control Panel

Industrial equipment data
infrastructure

■ The problem

Most field problems start with missing data

ISSUE 01

**Equipment stops at dawn,
and no one knows why**

Crews respond only after a failure, and lost production grows with every hour of downtime.

ISSUE 02

**When experts retire, their
know-how leaves too**

Judgment based on sound and experience is never recorded.

ISSUE 03

**Overheating and overload
go unrecorded**

Without a history of temperature, current and vibration, prevention is difficult.

The Intelligent Control Panel records data while it controls, and flags anomalies early.

The control panel becomes data infrastructure



Intelligent Control Panel test bed (illustration)

01

Real-time PLC control and data collection

Control logic and interlocks stay with the on-site PLC.

02

Direct to cloud, no extra hardware

Encrypted MQTT/TLS from the PLC, no gateway or industrial PC

03

Standardized data in the DataHub

Equipment → sensor → value → status, stored as time series

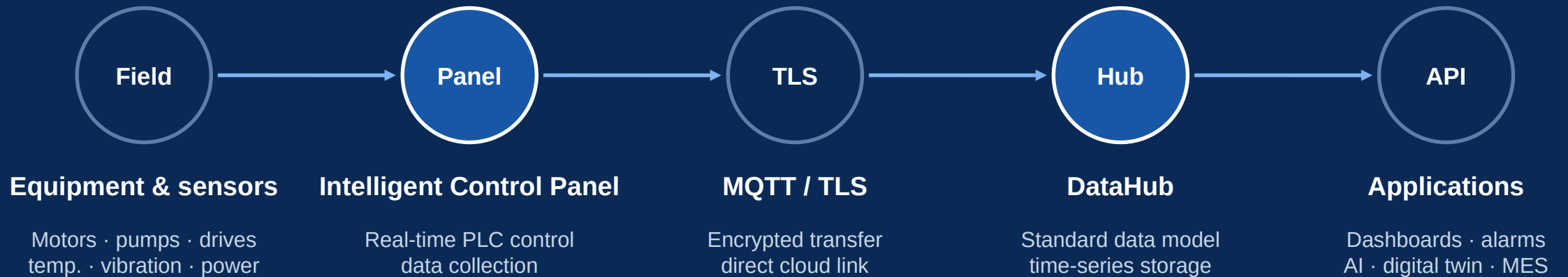
04

Anomaly alarms, expandable to PdM

Step-by-step upgrades with AI analysis and digital twins

■ How it works

From field equipment to AI, one continuous chain



Control stays on site

The server never controls equipment directly; interlocks and final control remain with the PLC.

Secure connection

Field and cloud are linked through TLS-encrypted MQTT.

A data model that scales

As equipment is added, AI and digital twins keep using the same standard data.

■ For manufacturers

Prepare for equipment failure with data, not guesswork

24/7 equipment records

Winding and bearing temperatures, current, power, vibration and speed collected automatically

Early anomaly alarms

Alarms when values leave the limits you set

Replacement and data in one step

One panel handles both control and data collection

Delivery process



Typical lead time 9–15 weeks, depending on site conditions.

■ For IT partners

Leave PLC integration and field work to us. Focus on your platform.

MODEL 01

OT scope delivery

We deliver panels, PLCs, site installation and commissioning in your projects.

MODEL 02

Data integration

Data collected by our panels, delivered in a standard structure via REST API and WebSocket.

MODEL 03

Joint proposals

We find manufacturing customers and programs together and propose jointly with demo equipment.

DONG-IN ENSIS

Field data · OT

- Panel design · build · commissioning
- PLC · drive · sensor integration
- Encrypted cloud transfer · DataHub
- Standard data model + API

IT partner

Platform · services

- MES · ERP · EAM integration
- AI analytics · predictive models
- Digital twins · dashboards
- Customer-facing platform

■ Applications

From land to sea, wherever motors run



Water booster pump station

160 kW × 3



Steel mill blower

250 kW



Air compressor room

90 kW × 2



Liquid hydrogen station

30 kW



LNG carrier engine room

75 kW × 2



Ballast water treatment

55 kW



Wind installation vessel crane

200 kW



Port container crane

300 kW

* Application scenarios. Images are illustrations.

Control technology proven in demanding sites

Design and build in-house

Yongsan Factories 1 & 2, dedicated FAT line, CE·UL-based design and build

Field experience

Control systems for hydrogen, shipbuilding and plant sites with strict safety requirements

R&D

Eco-Technology Research Center, national R&D, 3 registered patents on LH2 station control

Field	Key references
Hydrogen	Liquid hydrogen station control systems for a major Korean energy company (multiple sites)
Green ships	FGSS test facility and heater control for a leading shipbuilder
Green ships	Hybrid electric propulsion automation
Gas · energy	Marine FGSS auxiliary pump and heater control; BOG compressor control
Industrial plants	Resin plant E&I overseas (Vietnam); chemical plant MCC · PLC
Machinery	Packaging machine control for a global OEM; steel line control

* Customer names are withheld.

Certifications and milestones

Certifications

ISO 9001 · 14001 · 45001	INNOBIZ · MAINBIZ · Venture	Corporate R&D center
Innovation Premier 1000 (2026)	National Representative Innovative Co.	Regional Specialized Legend 50+
Busan Strategic Industry Leader	Hydrogen Specialized Co. (2026)	3 registered patents (LH2 station control)

Milestones

2026	Intelligent Control Panel business; Innovation Premier 1000; Hydrogen Specialized Co.; 3 patents
2025	ISO 45001; MAINBIZ · INNOBIZ
2024	Busan Strategic Industry Leader; Legend 50+
2023	Yangsan Factory 2; hydrogen vessel R&D; renamed DONG-IN ENSIS
2022	National Representative Innovative Company
2020	Eco-Technology Research Center founded
2006	Industrial and marine control systems business
1991	Founded as Dongin Gijeon



VISION

**Accelerating DX and AX in
manufacturing
and maritime with field data.**

MISSION

**Creating and connecting reliable data
from the panels that run the equipment.**

■ Contact us

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